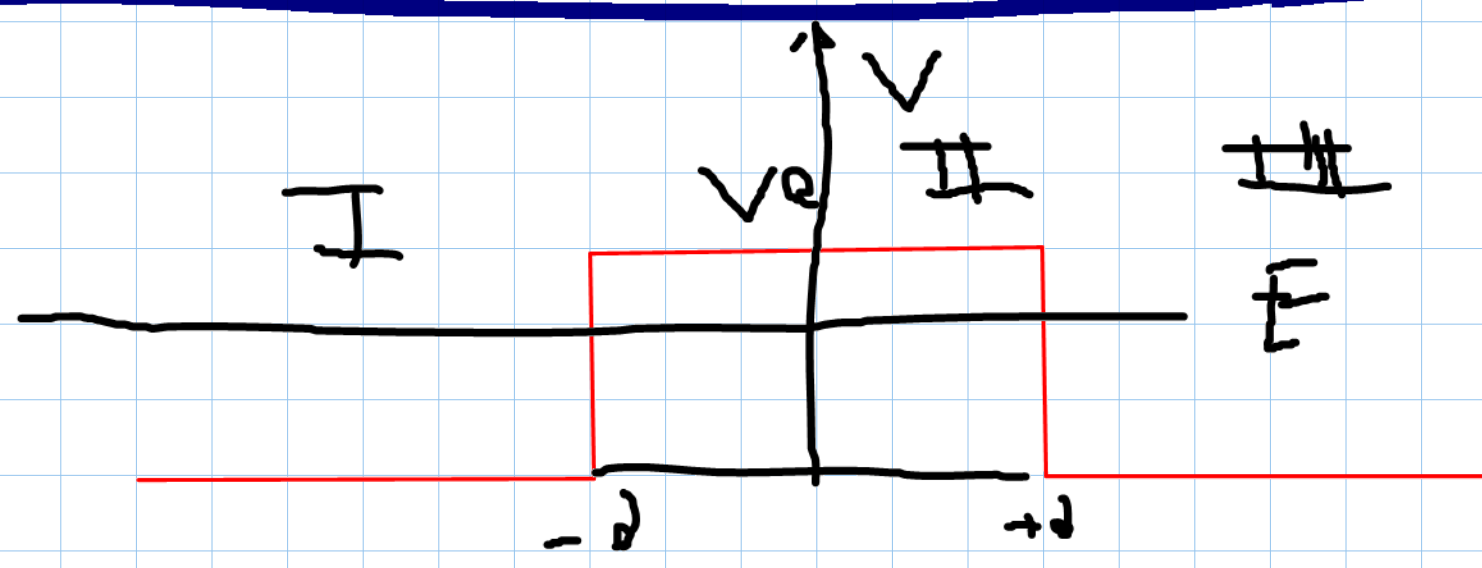


Quantum Mekaniği 5. Bölüm 3. Kısım

Tekrar



$$E < |V_0| \quad \text{Sch} \quad -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V\psi = E\psi (=)$$

Region I ve III benzer çözüm II farklı

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_I}{dx^2} = E\psi_I \Rightarrow \frac{d^2 \psi_I}{dx^2} = -\frac{2mE}{\hbar^2} \psi_I \Rightarrow \frac{d^2 \psi_I}{dx^2} + \underbrace{\frac{2mE}{\hbar^2}}_{k^2} \psi_I = 0$$

yani

I ve III için çözüm tipi

$Ae^{ikx} + Be^{-ikx}$ gibi olacak.

$$1e^{ikx} + Re^{-ikx}$$

Region 1
Region 3

$$0^2 + k^2 = 0 \quad \text{D = } \pm ik$$

Region için a.

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{II}}{dx^2} + V\psi_{II} = E\psi_{II} \quad E < |V|$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{II}}{dx^2} = (E - V)\psi_{II} \Rightarrow \frac{d^2 \psi_{II}}{dx^2} + (E - V)\psi_{II} = 0$$

V-E formatıyla yazılması lazım (Potansiyel kaydırma farklı) olması için

$$\frac{d^2 \psi_{II}}{dx^2} - \underbrace{\frac{2m(V-E)}{\hbar^2}}_K \psi_{II}(x) = 0$$

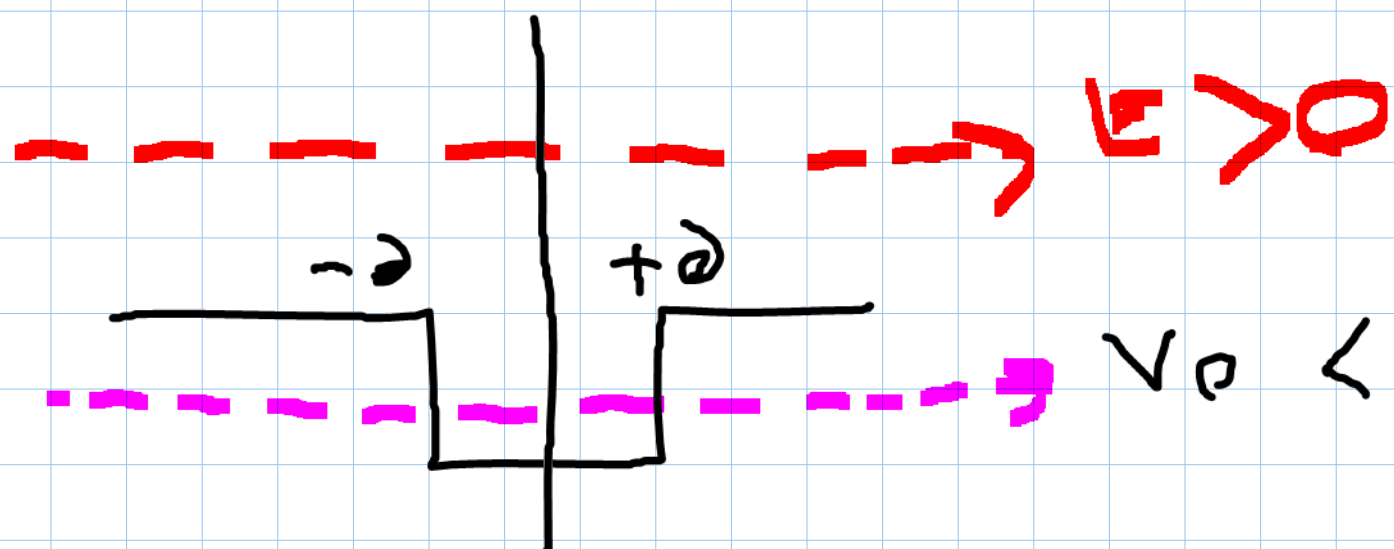
$$0^2 - K^2 = 0 \Rightarrow D = \pm K$$

$$R = \frac{ie^{-2ika} (K^2 - k^2) \sin(-2ika)}{-2ikK \cos(-2ika) - i(K^2 + k^2) \sin(-2ika)}$$

$$T = \frac{e^{-2ika} 2kK}{2kK}$$

Pot Bas
Kıy-
Engel ✓

Bir Potansiyel Kuyuda Bağlı Durumlar



$$V_0 < E < 0$$

$$E = -|E|$$

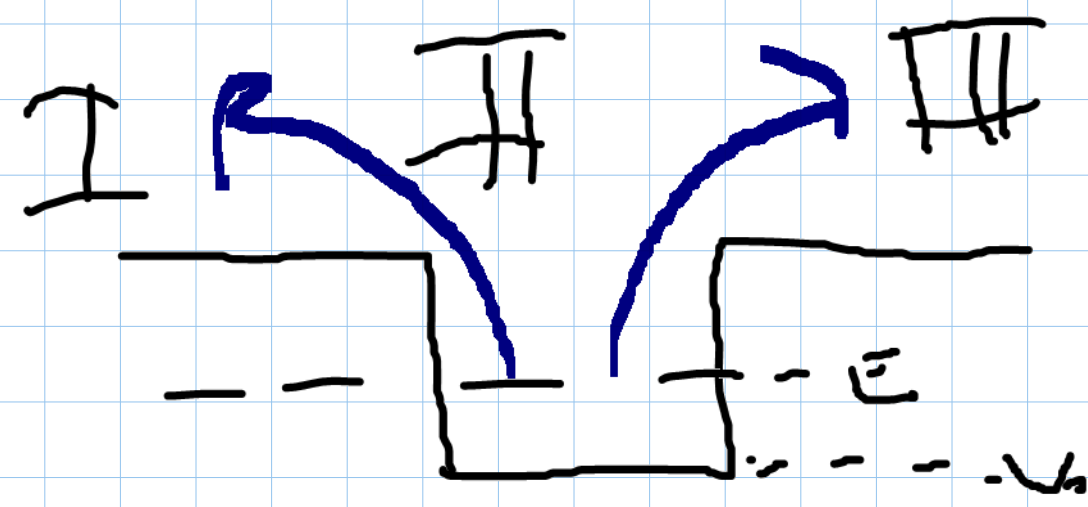
$$-\frac{\hbar^2}{2m} \frac{d^2 U}{dx^2} + VU = -|E|U$$

I ve III burada $\pm k$ lı sıklar

$$\text{I ve III } V = 0$$

$$\Rightarrow \frac{d^2 U}{dx^2} - \frac{2mE}{\hbar^2} U = 0 \Rightarrow D^2 - K^2 = 0 \quad D = \pm K$$

II için



$$-\frac{\hbar^2}{2m} \frac{d^2 U_{II}}{dx^2} - V_0 U_{II} = -|E| U_{II}$$

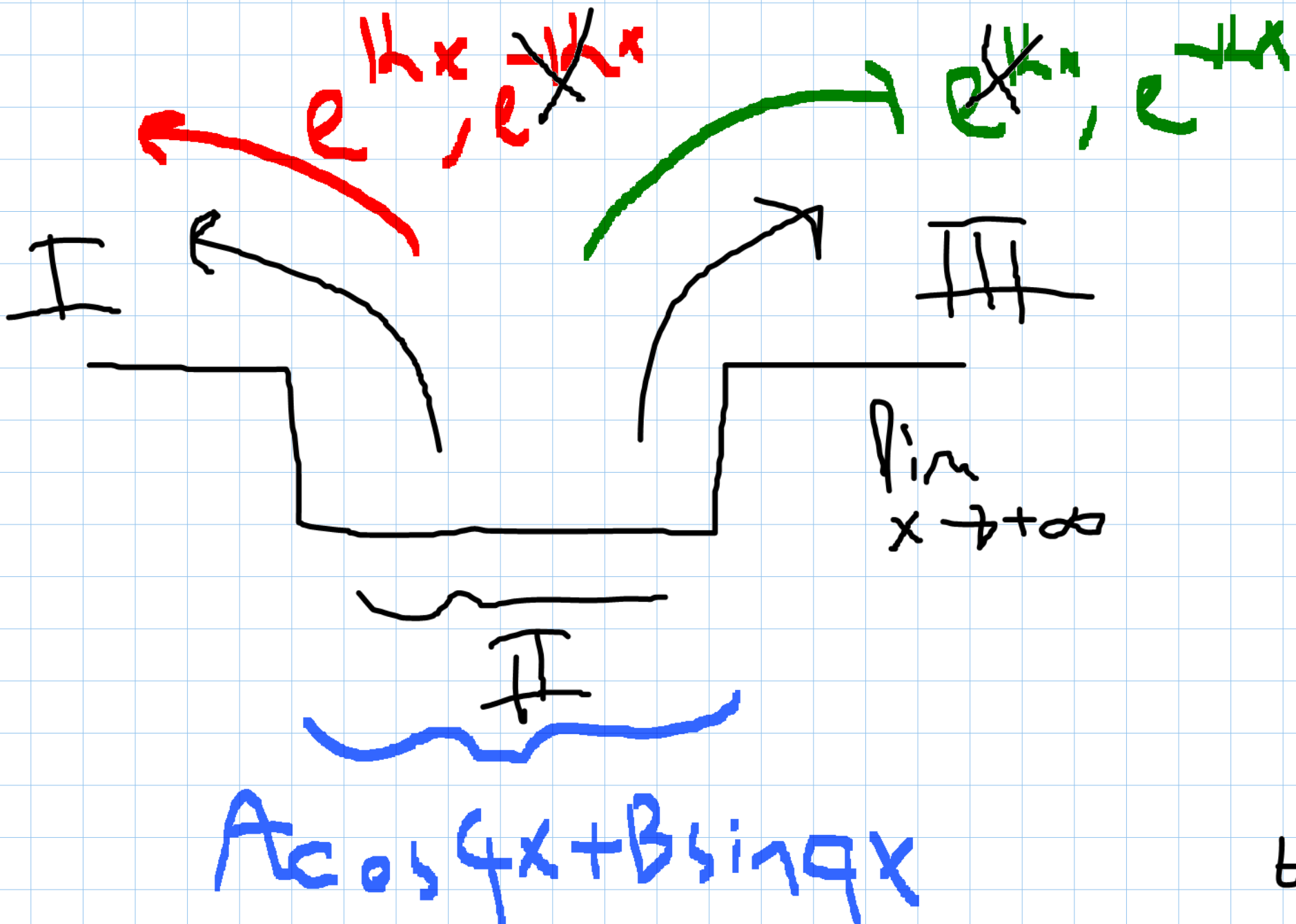
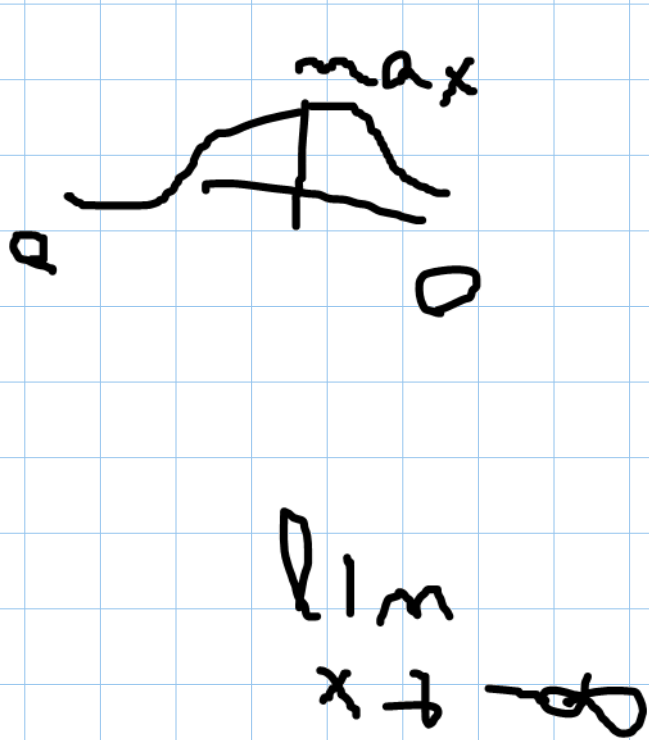
V_0 , olmak gerektiğinden $V_0 = |E|$ dir

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 U_{II}}{dx^2} = (V_0 - |E|) U_{II} \Rightarrow \frac{d^2 U_{II}}{dx^2} + \frac{2m(V_0 - |E|)}{\hbar^2} U_{II} = 0$$

$$D^2 + q^2 = 0$$

$$D = \pm iq$$

- 1) $e^{\pm iqx}$
- 2) $A \cos qx + B \sin qx$



$$\text{I ve III } x > a \quad C_1 e^{-Kx}$$

$$x < -a \quad C_2 e^{Kx}$$

$$\text{I ve III } \Rightarrow x > a \quad -K C_1 e^{-Kx}$$

$$x < -a \quad K C_2 e^{Kx}$$

$$\text{II için } A \cos qx + B \sin qx$$

$$\text{türevi } -Aq \sin qx + Bq \cos qx$$

$$u_{\pm}|_{-\infty} = u_{\pm}|_{-\infty}$$

$$u_{\pm}|_{\infty} = u_{\pm}|_{\infty}$$

$$u_{\pm}|_{-\infty} = u_{\pm}|_{-\infty}$$

$$u_{\pm}|_{\infty} = u_{\pm}|_{\infty}$$

$$x = -a \Rightarrow C_1 e^{-kx} = A \cos qa + B \sin qa \quad \left| \quad A \cos qa + B \sin qa = C_2 e^{kx} \right.$$

$$\Rightarrow +k C_1 e^{-kx} = -q A \sin qa + q B \cos qa \quad \left| \quad -q A \sin qa + q B \cos qa = -k C_2 e^{kx} \right.$$

$$k = q(A \sin qa + B \cos qa) / (A \cos qa - B \sin qa)$$

$$k = q(A \sin qa - B \cos qa) / (A \cos qa + B \sin qa)$$

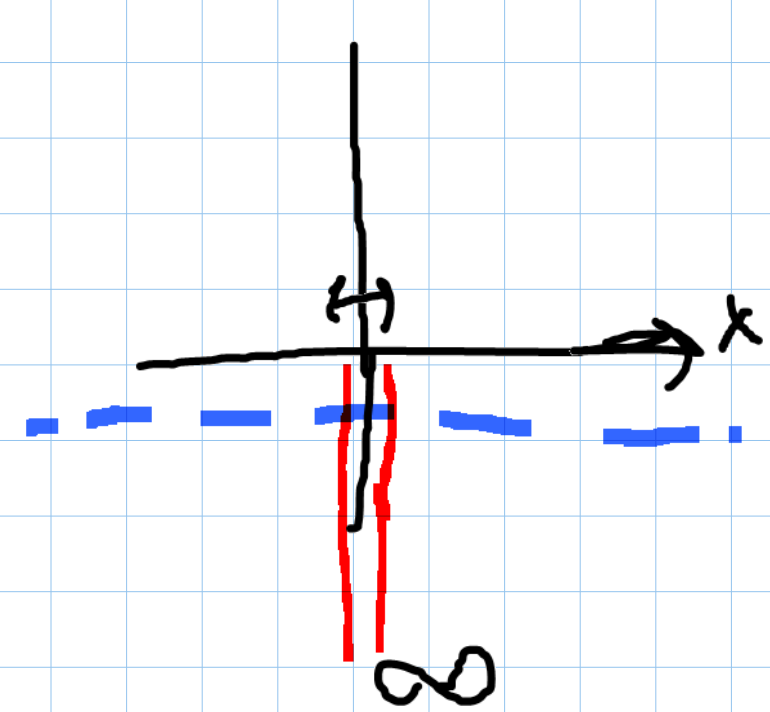
$$k \quad B=0 \text{ sa } k = q \tan qa \quad (\text{cift})$$

$$A=0 \text{ sa } k = -q \cot qa \quad (\text{tek})$$

buradan grafikler ve lambda(λ) bulunur.

Delta Fonks Pot

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} - V_0 \delta(x) u = -E u$$



$$I \text{ ve } II \quad V=0 \Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} = -E u$$

$$\frac{d^2 u}{dx^2} - \frac{2m|E|}{\hbar^2} u = 0$$

$$D_{1,2} = k^2 \quad e^k, e^{-k}$$

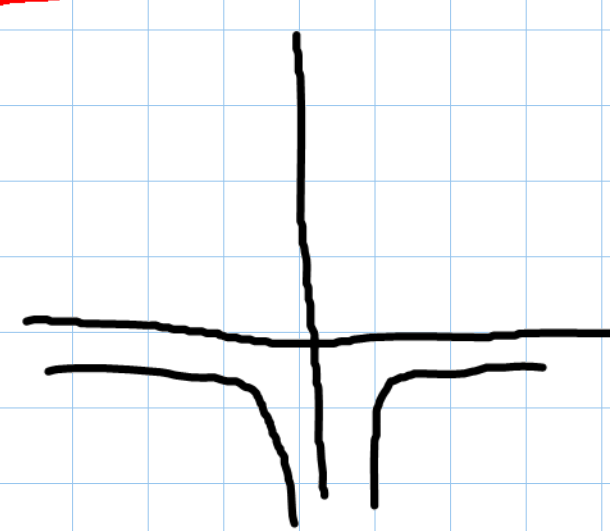
$$\text{sol} \quad \lim_{x \rightarrow -\infty} e^{kx} = 0 \quad V$$

$$\text{sag} \quad \lim_{x \rightarrow \infty} e^{-kx} = 0 \quad V$$

$x=0$ da fonklar sürekli olmalı

$$\text{2. m. türevleri: } \frac{du}{dx} \Big|_0^+ - \frac{du}{dx} \Big|_0^- = \frac{2m|V_0|}{\hbar^2} u(0)$$

Teklikenin 5 Fonks

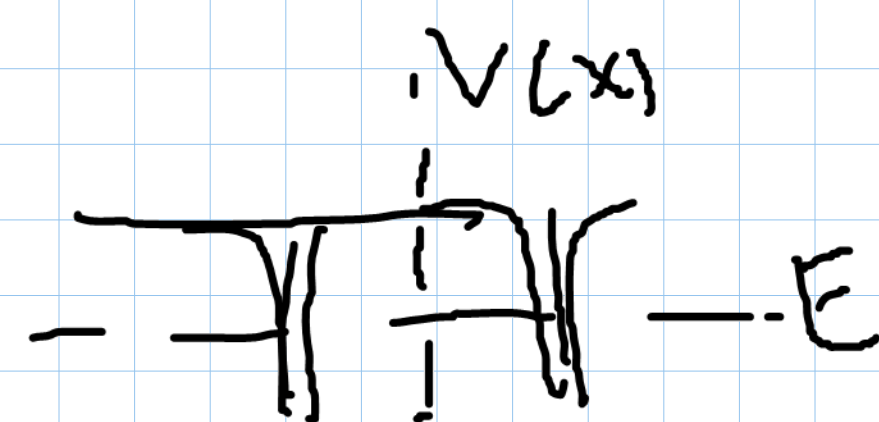


$$k = \frac{n}{2a}$$

$$k^2 = \frac{2mE}{\hbar^2} = \frac{n^2}{4a^2} \Rightarrow |E| = \frac{\hbar^2}{8ma^2}$$

Td Bağlı Durum

GK+ Kuyu



Kronig Penney

Videosu var



iletkenlik Band yapısı

→ Sonuç Grafik

